IAT New Test 12

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

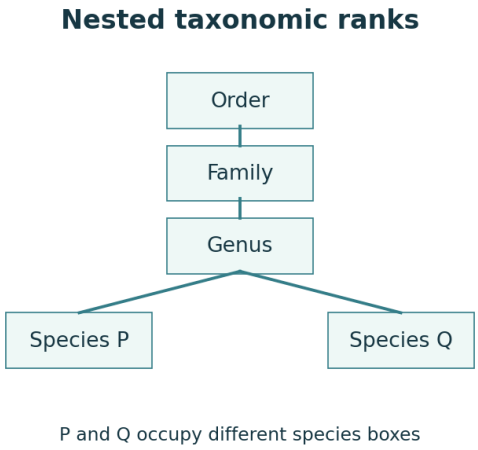
This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

Instructions

Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

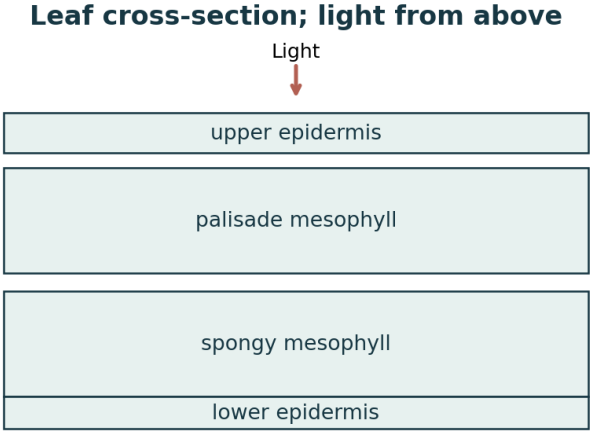
Biology

Q1 In the pictured hierarchy, two organisms belong to the same genus but different species. Which is the narrowest rank they necessarily share?



A) Genus
B) Family only
C) Order only
D) Species

Q2 The section shows palisade mesophyll immediately beneath the upper epidermis and spongy mesophyll below. Light arrives from above. Which comparison is best supported?

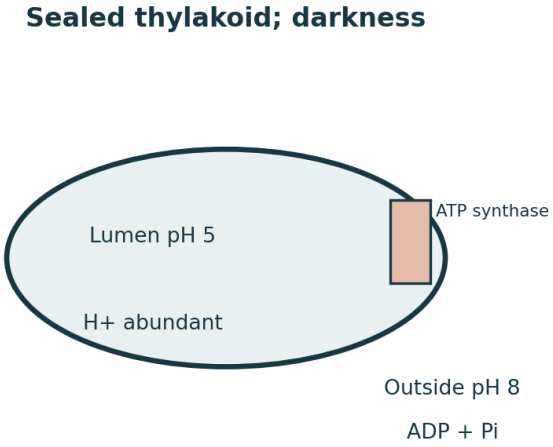


A) Stomata can occur only on the upper epidermis
B) Upper palisade cells receive incident light before the deeper spongy layer
C) The spongy layer contains no chloroplasts
D) All mesophyll cells have identical exposure

Q3 A homogenate is centrifuged sequentially: intact nuclei sediment first; a later, higher-speed pellet contains double-membraned organelles with their own circular DNA. Which activity is most enriched in that later pellet?

A) Transcription of nuclear chromatin
B) Photosynthetic light reaction in every animal cell
C) Aerobic ATP synthesis in mitochondria
D) Translation on free cytosolic ribosomes

Q4 A sealed thylakoid is kept in darkness. The lumen is pH 5 and the external medium pH 8; ADP and phosphate are outside, and the ATP-synthase head faces outward. What can initially occur if the membrane is proton-tight except through synthase?



A) Oxygen evolution without light
B) CO_2 fixation without ATP or NADPH
C) Protons move from pH 8 to pH 5 through synthase
D) Protons exit the lumen and drive ATP production outside

Q5 A transmitter is released normally, but a drug occupies postsynaptic receptors without opening the ion channel. Which observation best isolates the drug's action?

A) Directly faster axonal conduction in the postsynaptic neuron
B) No presynaptic action potential and no vesicle fusion
C) Higher postsynaptic potential with unchanged release
D) Normal presynaptic vesicle fusion but reduced postsynaptic potential

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Q6 A single megaspore mother cell forms four megaspores; only one survives and undergoes three mitoses before cellularisation. How many nuclei are present immediately before cellularisation in the surviving embryo sac?

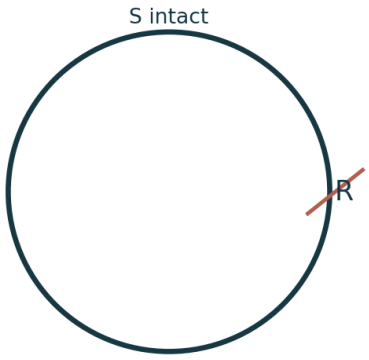
A) Eight
B) Four
C) Sixteen
D) Thirty-two

Q7 An autosomal recessive disease affects a child's sibling; both parents are unaffected obligate carriers. The child is unaffected. If this child has a partner known to be a carrier, what is the chance their first child is affected?
A) 1/4
B) 1/3
C) 1/6
D) 2/3

Q8 A rare resistance allele already exists in a bacterial culture before antibiotic is applied. Resistant colonies dominate afterward. Which inference is justified?
A) The antibiotic preferentially selected pre-existing resistant cells
B) The antibiotic intentionally created resistance in each survivor
C) All survivors acquired the allele from the drug molecule
D) Every initially sensitive bacterium must have mutated

Q9 The plasmid diagram marks a resistance gene R interrupted by the cloning site and an intact resistance gene S. A colony grows on S-antibiotic but fails on R-antibiotic after transformation. What is the most direct inference?

Plasmid markers and insertion site



Marked cloning site in R

A) The plasmid was absent
B) The inserted DNA disrupted S
C) Both markers were deleted
D) An insert likely interrupted R while S remained functional

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Q10 Two species consume the same limiting resource. Alone they persist at equal density; together A persists while B declines, but supplying abundant resource restores B. Which explanation best fits both observations?
A) A directly poisons B under all conditions
B) Predation by A is proved because B declines
C) B is genetically incapable of reproduction
D) Resource competition suppresses B when the shared resource is limiting

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Q11 A coupling-phase heterozygote AB/ab is test-crossed, giving 390 AB, 390 ab, 110 Ab and 110 aB progeny. Assume these are exact gamete probabilities, the same in both sexes, with no viability differences. Two such heterozygotes are then crossed. What is the probability of an offspring with genotype aa and dominant B phenotype?
A) 121/10000
B) 979/10000
C) 1521/10000
D) 2500/10000

Q12 An X-linked recessive allele is present in a man whose partner is unaffected and homozygous normal. Which child class is obligatorily a carrier under ordinary XX/XY inheritance?
A) All sons
B) Half the sons
C) All daughters
D) Half the daughters

Q13 Thyroid hormone rises above its normal set point in a person with an intact hypothalamic-pituitary axis. What immediate feedback change is expected?
A) TRH and TSH decrease through negative feedback
B) TSH secretion increases to reinforce the rise
C) Pituitary stops responding permanently
D) Only adrenal ACTH decreases

Q14 At identical substrate concentration, enzyme velocity falls after adding compound X. Raising substrate concentration restores the original maximum rate, while the concentration required for half-maximal speed increases. Which model fits?
A) Irreversible denaturation
B) Noncompetitive inhibition lowering maximum rate
C) Competitive inhibition at the substrate site
D) Loss of enzyme molecules from the mixture

Q15 A leaf's photosynthetic rate rises with light up to a plateau at fixed CO_2. Increasing CO_2 at the plateau raises the rate, but only until a new plateau. What does this demonstrate?
A) Light can never limit photosynthesis
B) CO_2 was limiting at the first plateau, though another factor later limits
C) CO_2 is irrelevant to carbon fixation
D) The same factor must limit at every plateau

Chemistry

Q16 The figure labels a carbon atom with one single C–C bond and one triple C≡N bond in CH₃–C≡N. What is the hybridisation of that carbon, and how many π bonds are incident on it?

Methyl cyanide bond geometry

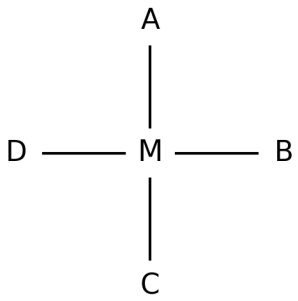


Single C–C; triple C≡N

A) sp^3, zero
B) sp^2, one
C) sp, one
D) sp, two

Q17 Four different monodentate ligands A, B, C and D occupy the four sites of a square-planar complex, as represented in the diagram. How many geometrical isomers are possible, counting arrangements related by rigid rotation as the same?

Four different ligands in a square plane



A, B, C and D denote four distinct monodentate ligands

A) 2
B) 4
C) 6
D) 3

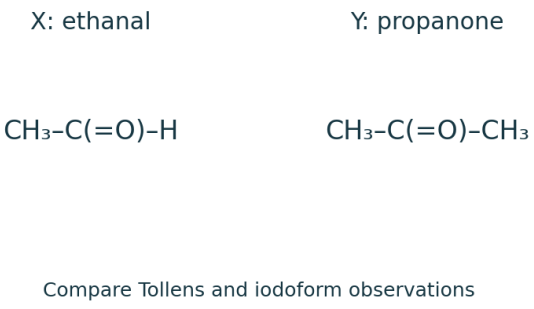
Q18 For nitrate, consider the octet-respecting Lewis contributors with one N=O and two N–O bonds. Which triple gives their number, the formal charge on nitrogen in each, and the mean N–O bond order in the resonance description?
A) 3; −1; 4/3
B) 3; +1; 4/3
C) 1; +1; 2
D) 3; 0; 5/3

Q19 Which proposed quantum-number set can describe an electron in an atomic orbital? Values are (n,l,m_l,m_s).
A) (3,3,0,+1/2)
B) (2,1,2,−1/2)
C) (4,2,−2,+1/2)
D) (1,0,0,+1)

Q20 Under otherwise comparable conditions, which carbocation is most stabilised by resonance?
A) $CH_3CH_2^+$
B) $(CH_3)_3C^+$
C) $CH_2{=}CH{-}CH_2^+$
D) CH_3^+

Q21 The two samples in the figure are ethanal (X) and propanone (Y). X gives Tollens silver; both X and Y give iodoform. Which conclusion follows?

Carbonyl structures in two test samples



A) Only X contains a methyl carbonyl group
B) Both samples are aldehydes
C) Y is an aldehyde because iodoform is positive
D) Tollens distinguishes X as an aldehyde whereas iodoform is not specific for aldehydes

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Q22 At equal concentration in water, aniline is less basic than cyclohexylamine. Which structural effect most directly explains this comparison?
A) Aniline nitrogen’s lone pair is delocalised into the aromatic ring
B) Cyclohexylamine’s lone pair is delocalised into an aromatic ring
C) Aniline has no nitrogen lone pair
D) Both amines are identical after protonation

Q23 Complete hydrolysis of the tripeptide Gly–Ala–Gly yields which products and mole ratio?
A) Gly:Ala=1:2
B) Gly:Ala=2:1
C) Gly:Ala=3:1
D) Only Gly, because terminal residues are removed

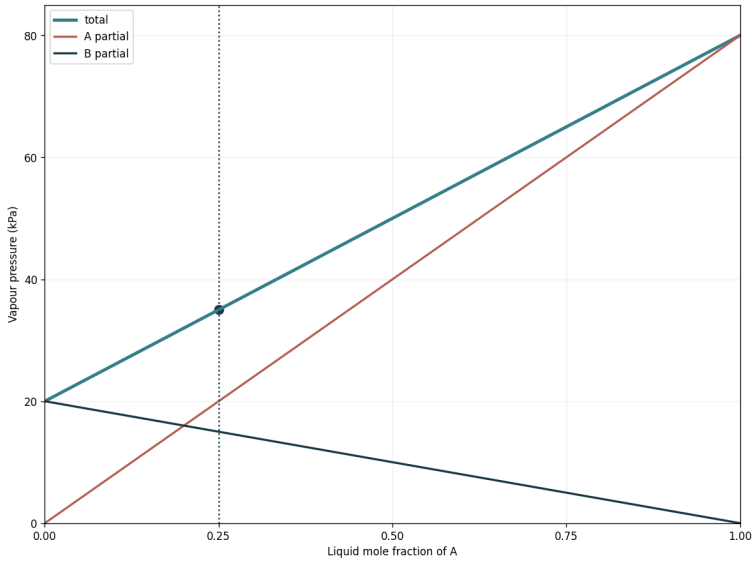
Q24 Initial-rate trials: [A],[B],rate = (0.10,0.10,2), (0.20,0.10,8), (0.10,0.20,4), with consistent units. What is the empirical rate law?
A) rate∝$[A]^2[B]$
B) rate∝$[A]^2[B]^2$
C) rate∝$[A][B]$
D) rate∝$[A][B]^2$

Q25 At 25°C, a 0.10 M NH_4Cl solution is prepared. Take $K_b(NH_3) = 10^{-5}$ and $K_w = 10^{-14}$; neglect activity effects and use the small-hydrolysis approximation. What is its approximate pH?
A) 3
B) 7
C) 9
D) 5

Q26 In acidic solution, MnO_4^- is reduced to Mn^{2+}. How many electrons does one permanganate ion accept in the balanced half-reaction?
A) Five
B) Three
C) Seven
D) Two

Q27 A reaction has ΔH>0 and ΔS>0, both approximately temperature independent in the relevant interval. Which temperature condition permits ΔG<0?
A) Only below ΔH/ΔS
B) Above ΔH/ΔS
C) At all temperatures
D) At no temperature

Q28 The figure gives ideal-mixture pure-component vapour pressures p*A=80 kPa and p*B=20 kPa. At liquid mole fraction xA=0.25, what is the total vapour pressure?



A) 20 kPa
B) 35 kPa
C) 50 kPa
D) 65 kPa

Q29 A 1.0 mol mixture of methane and ethane undergoes complete combustion and produces 1.4 mol CO_2. What mole fraction of the original mixture was ethane?
A) 0.4
B) 0.2
C) 0.6
D) 0.8

Q30 An optically pure 2-bromobutane undergoes clean SN2 replacement by iodide, then clean SN2 replacement by cyanide supplied as KCN. Assume carbon-bonded cyanide product, no elimination, no rearrangement and no other reaction. What are the final product's carbon count and net local geometry at the original stereogenic carbon relative to the starting material?
A) 4 carbons; retained
B) 5 carbons; inverted
C) 5 carbons; retained
D) 4 carbons; racemised

Mathematics

Q31 For A=[[1,2],[2,5]], solve $A(x,y)^T=(3,7)^T$. What is x+y?
A) 2
B) 1
C) 4
D) 3

Q32 Among integers 1 through 120, how many are divisible by exactly two of 2, 3 and 5?
A) 28
B) 40
C) 36
D) 24

Q33 On integers define aRb when a−b is divisible by 4. Which statement is true about R?
A) Reflexive and symmetric but not transitive
B) An equivalence relation with four residue classes
C) Transitive but not reflexive
D) Neither symmetric nor transitive

Q34 The cube roots of unity are plotted on the unit complex circle. What is the area of the triangle formed by them?
A) $3\sqrt{3}/2$
B) $\sqrt{3}/2$
C) $\sqrt{3}/4$
D) $3\sqrt{3}/4$

Q35 Evaluate $\int_0^2 x\sqrt{(4-x^2)}\,dx$.
A) 4/3
B) 2
C) 8/3
D) 16/3

Q36 The planes x+y+z=3 and x−y+z=1 intersect in a line. Which point lies on this line?
A) (1,1,1)
B) (0,1,1)
C) (2,0,0)
D) (0,2,1)

Q37 For $f_a(x)=x^3-3ax$ on $x\geq 1$, which real a makes f_a strictly increasing throughout its domain and makes its minimum value on the domain at most −2?
A) a>1
B) a=1
C) $a\leq 1$
D) No real a

Q38 Three observations have mean 4 and population variance 8. Two observations are 2 and 4. What is the third observation, and is the stated variance consistent?
A) 6, consistent
B) 8, consistent
C) 6, inconsistent
D) 8, inconsistent

Q39 For f(x)=(x²−9)/(x−3) when x≠3, what value of f(3) makes the function continuous?
A) 0
B) 3
C) 9
D) 6

Q40 A fair coin is tossed five times. What is the probability of exactly three heads given that the first toss was a head?
A) 1/4
B) 5/16
C) 3/8
D) 1/2

Q41 Find the sum of all distinct x in the closed interval [0,2π] satisfying sin x+cos x=1.
A) π/2
B) 5π/2
C) 2π
D) 3π/2

Q42 For x>0, solve y′+(1/x)y=x with y(1)=0. What is y(2)?
A) 7/6
B) 7/3
C) 1/6
D) 4/3

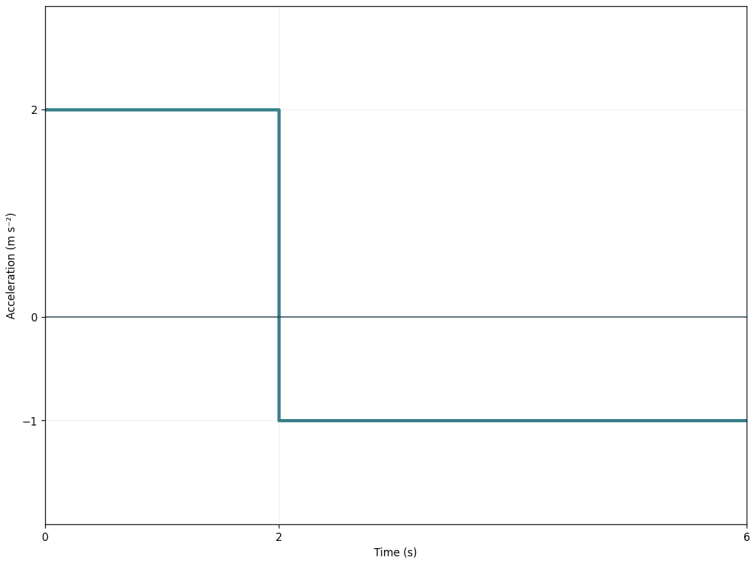
Q43 For which real k does $x^2 - 2x + k = 0$ have real positive roots α, β, not necessarily distinct, satisfying $\alpha^2 + \beta^2 \leq 3$?
A) $0 < k < 1$
B) $k \geq 1/2$
C) $1/2 \leq k < 1$
D) $1/2 \leq k \leq 1$

Q44 At the point (1,2) on y²=4x, what is the slope of the normal?
A) −2
B) 2
C) −1
D) 1

Q45 For a=(2,−1,2) and b=(1,2,2), what is the scalar projection of a onto b?
A) 4
B) 4/3
C) 8/3
D) 3

Physics

Q46 The acceleration-time graph shows a=+2 m s⁻² from 0 to 2 s and a=−1 m s⁻² from 2 to 6 s. A particle begins at 3 m s⁻¹. What is v at 6 s?



A) 3 m s⁻¹
B) 1 m s⁻¹
C) −1 m s⁻¹
D) 7 m s⁻¹

Q47 A 0.5 kg block descends 0.4 m without friction and compresses a horizontal spring of k=100 N m⁻¹. What is its maximum compression? Take g=10 m s⁻² and neglect height change during compression.
A) 0.10 m
B) 0.20 m
C) 0.40 m
D) 0.50 m

Q48 The same constant torque acts on two rigid rotors with moments of inertia I and 4I. What is the ratio of their angular accelerations α_1/α_2?
A) 1/4
B) 1/2
C) 4
D) 16

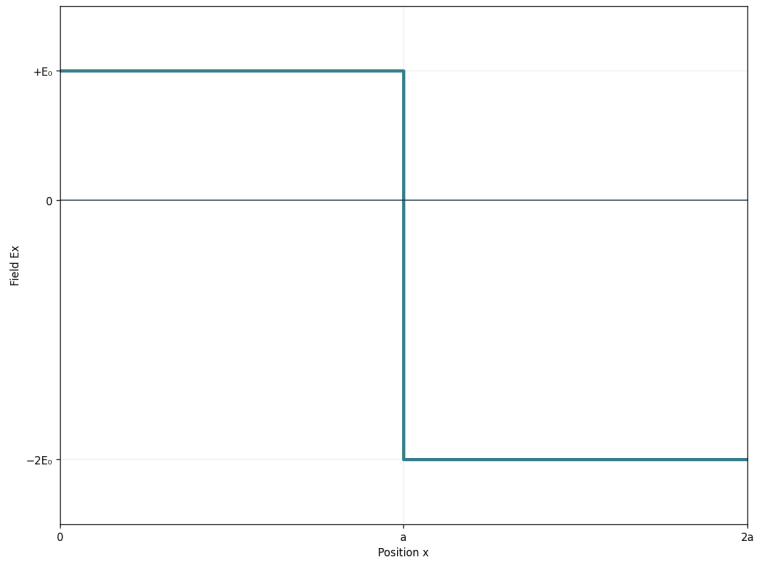
Q49 Small spheres A and B fall at terminal speed in the same viscous fluid of density ρ. A has radius r and density 3ρ; B has radius 2r and density 2ρ. Assume Stokes drag, low Reynolds number and no boundary effects. What are ratios v_B/v_A and P_B/P_A, where P is the power dissipated by viscous drag?
A) 4 and 32
B) 2 and 4
C) 4 and 8
D) 2 and 8

Q50 An ideal gas expands adiabatically and reversibly from state 1 to state 2. Which combination is correct for work done by the gas, internal-energy change and heat transfer?
A) Positive work, negative ΔU, zero heat
B) Positive work, positive ΔU, zero heat
C) Zero work, zero ΔU, positive heat
D) Negative work, negative ΔU, positive heat

Q51 A tuning fork makes 4 beats per second with a 250 Hz reference. When the reference frequency is increased to 252 Hz, beat frequency falls to 2 Hz. What is the fork frequency?
A) 246 Hz
B) 254 Hz
C) 248 Hz
D) 252 Hz

Q52 A satellite changes from a circular orbit of radius r to a circular orbit of radius 4r about the same planet. What is the new orbital period divided by the old one?
A) 2
B) 4
C) 8
D) 16

Q53 The graph gives Ex=+E_0 between x=0 and a and Ex=−2E_0 between x=a and 2a. What is V(2a)−V(0)?



A) +E_0a
B) −E_0a
C) −3E_0a
D) +3E_0a

Q54 A parallel-plate capacitor stays connected to an ideal battery of fixed voltage V while its gap is slowly doubled, with area unchanged. Let its initial field energy be U. Neglect heating and fringing. What are the work done by the external agent moving the plates and the work done by the battery, respectively?
A) $-U/2;\ +U$
B) $+U;\ -U/2$
C) $0;\ -U/2$
D) $+U/2;\ -U$

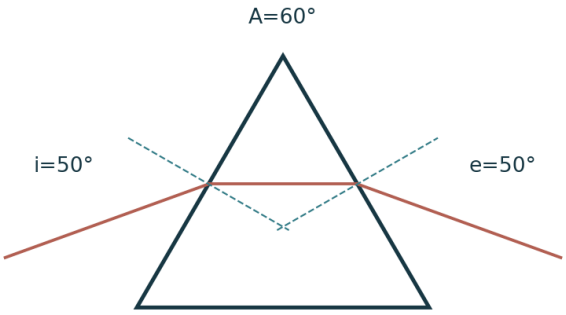
Q55 A 12 V cell with internal resistance 1 Ω drives an external 5 Ω resistor. What is its terminal voltage under load?
A) 14 V
B) 12 V
C) 2 V
D) 10 V

Q56 Two long parallel wires carry steady currents in the same direction. If the separation doubles but currents remain unchanged, what happens to force per unit length?
A) It doubles
B) It halves, and the force remains attractive
C) It quarters and becomes repulsive
D) It stays unchanged

Q57 In a series RLC circuit driven at variable frequency, the resistance is increased while L and C stay fixed. Which change is expected near resonance in the current-versus-frequency curve?
A) A sharper and taller peak
B) The resonance frequency necessarily doubles
C) A lower, broader current peak
D) Zero current at resonance

Q58 The prism sketch shows a ray passing symmetrically: angle of incidence equals angle of emergence and the prism apex angle is 60°. If incidence is 50°, what is the deviation?

Symmetric ray path through a prism



A) 20°
B) 40°
C) 30°
D) 50°

Q59 Two photons have wavelengths 400 nm and 600 nm in vacuum. What is the ratio f_{400}/f_{600}?
A) 2/3
B) 1
C) 3/2
D) 9/4

Q60 Two independent radioactive isotopes A and B start with equal activities. Their half-lives are 1 h and 2 h respectively. What is the ratio of total activity after 2 h to initial total activity?

A) 1/4
B) 1/2
C) 5/8
D) 3/8

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / A taxonomic hierarchy

Answer: (A) Genus.

Why: Genus encloses multiple related species. Membership in one genus entails shared family and higher ranks, but the question asks for the narrowest guaranteed shared rank. Distinct species excludes species as a common assignment.

Why not the others: (B) family is shared but is not the narrowest known rank; (C) order is broader still; (D) species is explicitly different for P and Q.

Key point: Move upward from species only as far as necessary.

Q2 / B leaf mesophyll distribution

Answer: (B) Upper palisade receives incident light first.

Why: The depicted vertical arrangement places the palisade layer between the illuminated surface and spongy tissue. It supports a light-exposure inference without claiming that deeper cells cannot photosynthesise.

Why not the others: (A) stomatal placement is not fixed by this section; (C) spongy cells also contain chloroplasts; (D) ignores the layer order.

Key point: Use the indicated geometry, not an invented absolute rule.

Q3 / C organelle fractionation

Answer: (C) Mitochondrial aerobic ATP synthesis.

Why: Nuclear material was removed in the first pellet, while the second fraction is consistent with mitochondria, which are double membraned and have their own DNA. Enrichment means higher relative concentration, not exclusive localization.

Why not the others: (A) follows nuclei; (B) animal cells lack chloroplasts; (D) free ribosomes largely remain in supernatant.

Key point: Link sedimentation order to organelle structure and function.

Q4 / D chemiosmotic gradient

Answer: (D) Lumen-to-outside proton flow can synthesize ATP.

Why: The lumen has greater hydrogen-ion concentration; a permitted path through ATP synthase releases stored gradient energy to phosphorylate external ADP. Darkness blocks a sustained photosynthetic gradient, but an imposed initial gradient can transiently drive ATP synthesis.

Why not the others: (A) requires light-driven water splitting; (B) ignores energy inputs; (C) reverses spontaneous diffusion.

Key point: Follow both concentration gradient and catalytic-head side.

Q5 / D neural synapse and receptor

Answer: (D) Normal release with weak postsynaptic response.

Why: The premise preserves presynaptic release, so an impaired electrical response after receptor occupancy localizes the defect to receptor-mediated transduction at the postsynaptic membrane. A control of release distinguishes this from upstream failure.

Why not the others: (A) concerns propagation rather than the tested synapse; (B) changes the presynaptic premise; (C) predicts enhancement.

Key point: Separate transmitter release from receptor response.

Q6 / A embryo-sac development

Answer: (A) Eight nuclei.

Why: Meiotic products are four separate megaspores, but only one functional megaspore continues. Each of its three mitotic rounds doubles nuclei: one becomes two, then four, then eight. Counting all original megaspores would overcount the developing sac.

Why not the others: (B) stops after two divisions; (C) adds an extra mitosis; (D) multiplies all four meiotic products.

Key point: Track only the functional megaspore after meiosis.

Q7 / C conditional pedigree risk

Answer: (C) 1/6.

Given: Carrier parents Aa × Aa; unaffected child excludes aa.

Formula/Principle: Conditional carrier chance is 2/3; two carriers produce aa with probability 1/4.

Step 1: (2/3)(1/4)=1/6.

Why: Restrict the parental offspring distribution to AA and Aa after learning the person is unaffected. The affected sibling establishes parental genotypes but does not make the unaffected child certainly a carrier.

Why not the others: (A) assumes carrier status; (B) uses wrong conditional fraction; (D) gives carrier risk, not affected-offspring risk.

Key point: Condition on unaffected status before mating probability.

Q8 / A antibiotic resistance selection

Answer: (A) Selection changed allele frequencies.

Why: Pre-existing heritable variation and differential survival suffice to explain enrichment of resistance. The drug is the selecting environment, and the observation alone does not establish that it induced mutations in all survivors.

Why not the others: (B) assumes the drug deliberately creates a useful mutation; (C) treats an antibiotic molecule as an allele donor; (D) assumes every sensitive cell changed rather than dying.

Key point: Selection acts on variation already present.

Q9 / D vector selection and marker

Answer: (D) Insertional inactivation of R.

Why: Growth with S-antibiotic supports the intact S marker and plasmid uptake; loss of R growth is consistent with a DNA insert disrupting R at the shown site. A control uncut vector would help exclude unrelated mutations.

Why not the others: (A) conflicts with S selection; (B) predicts loss of S growth; (C) predicts loss on both media.

Key point: Read each selection phenotype against its own gene.

Q10 / D competition and resource

Answer: (D) Resource-mediated competition.

Why: The abundance intervention removes the limiting-resource condition and rescues B, providing evidence that resource depletion, rather than unavoidable toxicity, drives its decline in the mixed culture. Equal monoculture performance does not imply equal competitive ability.

Why not the others: (A) would persist after supplementation; (B) does not follow from resource rescue; (C) contradicts B alone.

Key point: A rescue treatment tests the proposed cause.

Q11 / B Linked test-cross gametes used in an intercross

Answer: (B) 979/10000.

Given: Gamete probabilities are $P(AB) = P(ab) = 0.39$ and $P(Ab) = P(aB) = 0.11$.

Formula/Principle: The desired event is aa excluding aabb; parental gametes combine independently under the stated model.

Step 1: Each parent supplies allele a with probability $0.39 + 0.11 = 0.50$, so

$$P(aa) = 0.50^2 = 0.25.$$

Step 2: Both parents must supply ab to produce aabb:

$$P(aabb) = 0.39^2 = 0.1521.$$

Step 3:

$$P(aaB_) = P(aa) - P(aabb) = 0.25 - 0.1521 = 0.0979.$$

Why: Linkage changes the joint gamete probabilities even though each single locus still segregates one half. Using the measured ab frequency preserves that dependence; multiplying independent-locus phenotype ratios would not.

Why not the others: (A) squares a recombinant gamete only; (C) reports aabb, the excluded class; (D) includes all aa offspring, including bb.

Key point: Infer gametes from the test cross before calculating a different cross.

Q12 / C sex-linked transmission

Answer: (C) All daughters.

Why: The father gives his sole X chromosome to every daughter and his Y to every son. Each daughter also receives a normal maternal X, so she carries the allele without expressing a recessive trait under the stipulated genotype.

Why not the others: (A) incorrectly sends a paternal X to every son; (B) assigns that X to half the sons despite Y inheritance; (D) overlooks that every daughter receives the father's single X.

Key point: Trace paternal X and Y separately.

Q13 / A endocrine feedback

Answer: (A) Lower TRH and TSH.

Why: Elevated thyroid hormone inhibits hypothalamic TRH and anterior-pituitary TSH secretion, reducing stimulation of the thyroid gland. This is a reversible homeostatic response in the stated intact axis.

Why not the others: (B) is positive feedback; (C) asserts permanent damage; (D) confuses endocrine axes.

Key point: Increased final hormone suppresses upstream drive.

Q14 / C enzyme inhibition

Answer: (C) Competitive inhibition.

Why: More substrate overcomes X, leaving the limiting maximum velocity unchanged but increasing apparent substrate requirement. The paired observations distinguish occupancy competition from reduced active enzyme amount.

Why not the others: (A) heat-denatured enzyme would not regain the same maximum with more substrate; (B) noncompetitive inhibition lowers maximum velocity; (D) fewer active enzyme molecules also lower maximum capacity.

Key point: Examine both limiting velocity and substrate dependence.

Q15 / B light-response limiting factor

Answer: (B) Limitation can shift.

Why: A response to increased CO_2 at unchanged light shows that CO_2 supply constrained the first plateau. The second plateau shows that additional CO_2 no longer changes the rate there; light, temperature, enzyme capacity or another factor may now constrain it.

Why not the others: (A) ignores that light can limit at low intensity; (C) contradicts the observed CO_2 response; (D) assumes the same bottleneck persists despite the intervention.

Key point: Diagnose the limiting factor by changing one input.

Chemistry

Q16 / D hybridisation and sigma-pi count

Answer: (D) sp with two π bonds.

Why: The nitrile carbon has two electron-domain directions: one σ bond toward methyl and one σ component toward nitrogen. Linear sp hybrid orbitals form those σ bonds; its two unhybridised p orbitals form the two π components of the triple bond.

Why not the others: (A) counts single bonds only; (B) treats a triple bond as a double; (C) misses one π component.

Key point: A triple bond counts as one domain but has two π bonds.

Q17 / D Square-planar trans-partner pairing with four distinct ligands

Answer: (D) Three geometrical isomers.

Given: Every ligand is different and all four sites lie in one square plane.

Formula/Principle: Fix A at one site by rotation and distinguish which ligand lies opposite it.

Step 1: The trans partner of A can be B, C or D.

Step 2: For each chosen trans partner, the remaining two ligands fill the other opposite pair; exchanging them does not create a new geometrical isomer after a rigid rotation in three-dimensional space.

Step 3: There are therefore three distinct trans-pairing patterns.

Why: A square-planar complex with four different ligands requires tracking pairings, not merely the two adjacent/opposite cases of MA_2B_2. The plane of the molecule also prevents counting an extra optical pair for each drawing.

Why not the others: (A) reuses the MA_2B_2 count; (B) counts a rotational duplicate; (C) counts the two orders of each remaining ligand pair separately.

Key point: Count independent opposite-partner pairings after removing rotational duplicates.

Q18 / B Nitrate resonance formal charge and bond-order bookkeeping

Answer: (B) Three contributors; nitrogen +1; mean bond order 4/3.

Formula/Principle: Formal charge equals valence electrons minus nonbonding electrons minus half the bonding electrons.

Step 1: The double bond can be placed on any of three oxygens, giving three equivalent contributors.

Step 2: Nitrogen has four bonding pairs and no lone pair, so its formal charge is $5-0-4=+1$. Two singly bound oxygens each carry -1, giving total -1.

Step 3: Each contributor has total bond order $2+1+1=4$ over three N–O bonds, giving mean $4/3$.

Worked derivation:

$$N_{\text{contributors}}=3, \quad \text{FC(N)}=5-0-8/2=+1, \quad \overline{\text{BO}}=(2+1+1)/3=4/3.$$

Why: Resonance distributes the double-bond character over equivalent bonds without giving nitrogen an expanded octet. Counting contributors, assigning formal charge and averaging bond order are separate operations and must all agree with the ion's total charge.

Why not the others: (A) assigns the ion's overall charge to nitrogen; (C) treats one contributor as the whole description; (D) miscounts both nitrogen's shared electrons and total bond order.

Key point: Check octet, total charge and resonance equivalence together.

Q19 / C quantum-number exclusions

Answer: (C) (4,2,−2,+1/2).

Why: For n=4, l can be 0 to 3, and l=2 permits m_l from −2 to +2; spin is ±1/2. All four numbers in the selected tuple meet these nested constraints.

Why not the others: (A) requires l<n; (B) has magnitude m_l greater than l; (D) uses spin +1 instead of ±1/2.

Key point: Check allowed ranges in sequence.

Q20 / C stability of reactive intermediates

Answer: (C) Allylic carbocation.

Why: The empty p orbital adjacent to the C=C bond can delocalise positive charge over the allylic framework. Alkyl substitution stabilises tertiary carbocations by other effects, but only the chosen structure has the requested resonance stabilisation.

Why not the others: (A) ethyl has no adjacent C=C π system; (B) tertiary alkyl substitution gives hyperconjugation rather than the specified resonance; (D) methyl has no neighboring π bond.

Key point: Distinguish resonance from hyperconjugation and induction.

Q21 / D Tollens versus iodoform evidence

Answer: (D) Different tests probe different structural features.

Why: Ethanal reduces Tollens reagent and has a CH_3CO– pattern; propanone has the methyl ketone pattern but is not an aldehyde, so it gives iodoform without Tollens silver under the stated test. Joint evidence separates carbonyl class from methyl-carbonyl response.

Why not the others: (A) wrongly excludes the methyl ketone Y from iodoform response; (B) treats a ketone as an aldehyde; (C) mistakes an iodoform-positive methyl ketone for an aldehyde.

Key point: A positive iodoform test does not uniquely identify an aldehyde.

Q22 / A basicity in solvent

Answer: (A) Aromatic lone-pair delocalisation.

Why: Delocalisation of aniline's unprotonated nitrogen lone pair reduces its availability to accept a proton. Cyclohexylamine lacks conjugation with an aromatic π system; the stipulated aqueous comparison is consistent with the difference.

Why not the others: (B) invents an aromatic ring; (C) contradicts nitrogen valence; (D) ignores different conjugate-acid structures and solvation.

Key point: Consider lone-pair availability in the specified solvent.

Q23 / B peptide hydrolysis products

Answer: (B) Two glycines per alanine.

Why: Hydrolysis breaks both peptide bonds without changing residue identity. The original sequence has glycine at both ends and alanine in the middle, yielding two moles of glycine and one mole of alanine per mole of peptide.

Why not the others: (A) reverses the counts; (C) ignores alanine; (D) treats bond cleavage as residue deletion.

Key point: Count residues before and after complete hydrolysis.

Q24 / A initial-rate order table

Answer: (A) Second order in A, first in B.

Why: Doubling A at fixed B multiplies the rate by four, giving exponent 2. Doubling B at fixed A multiplies the rate by two, giving exponent 1. Initial rates establish an empirical law, not necessarily the molecularity of an elementary step.

Why not the others: (B) predicts fourfold for B; (C) predicts twofold for A; (D) reverses the observed exponents.

Key point: Compare one concentration change at a time.

Q25 / D Conjugate-acid hydrolysis of an ammonium salt

Answer: (D) Approximately 5.

Given: Chloride is a spectator, while ammonium is the conjugate acid of ammonia.

Formula/Principle:

$$K_a(\mathrm{NH_4^+}) = K_w/K_b(\mathrm{NH_3}), \qquad [\mathrm{H^+}] \simeq \sqrt{K_a C}.$$

Step 1:

$$K_a = 10^{-14}/10^{-5} = 10^{-9}.$$

Step 2:

$$[\mathrm{H^+}] \simeq \sqrt{10^{-9}(0.10)} = 10^{-5}\ \mathrm{M}, \quad pH = 5.$$

Step 3: The hydrolysed fraction is $10^{-5}/0.10 = 10^{-4}$, supporting the small-hydrolysis approximation.

Why: A salt of a weak base and strong acid can be acidic because its cation donates a proton to water. The base's K_b cannot be used directly as the ammonium acid constant.

Why not the others: (A) uses the wrong acid constant; (B) assumes every salt is neutral; (C) treats ammonium as a base rather than its conjugate acid.

Key point: Identify the hydrolysing ion and its conjugate equilibrium.

Q26 / A half-reaction electron balance

Answer: (A) Five electrons.

Why: Mn changes oxidation state from +7 in MnO_4^- to +2 in Mn^{2+}, so its reduction requires five electrons. Balancing oxygen as water and hydrogen as H^+ does not change that electron count: $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$.

Why not the others: (B) uses a three-electron count without the Mn oxidation-state difference; (C) imports the +7 oxidation state as the electron count; (D) uses the final +2 state instead of the difference.

Key point: Electron count equals the central atom's oxidation-state decrease.

Q27 / B enthalpy versus entropy sign

Answer: (B) Above the threshold.

Why: $\Delta G = \Delta H - T\Delta S$. Both terms are positive initially, but the entropy term grows with temperature; ΔG becomes negative when $T\Delta S$ exceeds ΔH, or $T > \Delta H/\Delta S$ in kelvin.

Why not the others: (A) reverses the inequality; (C) ignores positive enthalpy at low T; (D) ignores increasing $T\Delta S$.

Key point: Solve the inequality with the sign of ΔS fixed.

Q28 / B partial-vapour-pressure table

Answer: (B) 35 kPa.

Given: xB=0.75 and the mixture follows Raoult's law.

Formula/Principle: p_total=xA p*A+xB p*B.

Step 1: 0.25×80+0.75×20=20+15=35 kPa.

Worked derivation:

$$p_{\text{total}} = x_A p_A^\circ + (1 - x_A) p_B^\circ = 0.25(80) + 0.75(20) = 35 \text{ kPa}.$$

Why: The total pressure sums component partial pressures, with liquid-phase mole fractions weighting the pure-component values. At xA=0.25, A contributes 20 kPa and B contributes 15 kPa, so neither pure-component pressure alone represents the mixture.

Why not the others: (A) uses B alone; (C) takes a simple average; (D) reverses the fractions.

Key point: Distinguish liquid composition from vapour pressure contribution.

Q29 / A combustion composition

Answer: (A) 0.4 ethane mole fraction.

Given: Methane contributes one CO_2 per molecule and ethane two.

Formula/Principle: Let ethane fraction be x; carbon balance gives (1−x)+2x=1.4.

Step 1: 1+x=1.4, hence x=0.4. Cross-check the product is between 1 and 2 mol.

Worked derivation:

$$n_{\text{CH}_4} = 1 - x, \quad n_{\text{C}_2\text{H}_6} = x,$$
$$n_{\text{CO}_2} = (1 - x) + 2x = 1.4,$$
$$x = 0.40.$$

Why: The extra 0.4 mol carbon dioxide over pure methane arises from the second carbon in each ethane molecule. Oxygen supplied in combustion does not change the carbon balance.

Why not the others: (B) would yield 1.2 mol CO_2; (C) yields 1.6; (D) yields 1.8.

Key point: Use product carbon balance to infer a mixture fraction.

Q30 / C Two stereospecific substitutions and nitrile carbon accounting

Answer: (C) Five carbons; net retention of local geometry.

Step 1: The first backside substitution replaces Br by I with inversion at the stereogenic carbon.

Step 2: The second backside substitution replaces I by carbon-bonded CN with another inversion. Two inversions restore the original local arrangement of the three unchanged groups relative to the entering group.

Step 3: Cyanide contributes one carbon to the original four-carbon skeleton, giving five carbons in the nitrile.

Why: Each clean SN2 step has its own stereochemical consequence, so the overall result cannot be inferred from one step alone. Net local retention is a geometric statement; an R/S label should still be assigned from the final substituent priorities rather than assumed from the word retention.

Why not the others: (A) omits the cyanide carbon; (B) counts only one inversion; (D) both omits the added carbon and invokes an SN1-style loss of stereospecificity absent from the premises.

Key point: Track bond formation and stereochemistry across every substitution.

Mathematics

Q31 / A matrix equation

Answer: (A) 2.

Given: x+2y=3 and 2x+5y=7.

Formula/Principle: Eliminate x using the first equation.

Step 1: Subtract twice the first from the second to obtain y=1; then x=1.

Worked derivation:

$$2x + 5y - 2(x + 2y) = 7 - 6 \Rightarrow y = 1, \quad x = 3 - 2 = 1, \quad x + y = 2.$$

Why: The elimination fixes both coordinates before their requested sum is taken. A common shortcut is to add the two right-hand-side values or read a single coordinate without checking that the pair satisfies both matrix rows.

Why not the others: (B) reports one coordinate; (C) adds one twice; (D) reports the original right-hand-side component.

Key point: Solve first, then calculate the requested expression.

Q32 / A inclusion-exclusion count

Answer: (A) 28 integers.

Given: The requested integers are divisible by exactly two of 2, 3 and 5 within 1 through 120.

Formula/Principle: Count pairwise intersections, then remove all appearances of the triple intersection.

Step 1: Pairwise counts are

$$\lfloor 120/6 \rfloor = 20, \quad \lfloor 120/10 \rfloor = 12, \quad \lfloor 120/15 \rfloor = 8.$$

Step 2: Triple divisibility means divisibility by 30, giving $\lfloor 120/30 \rfloor = 4$.

Step 3: Each exactly-two integer appears once in the pairwise sum, but each triple-divisible integer appears three times and must disappear entirely.

Step 4:

$$N = 20 + 12 + 8 - 3(4) = 28.$$

Worked derivation:

$$N_2 = \sum_{\text{three pairs}} N_{\text{pair}} - 3N_{\text{triple}} = 28.$$

Why: This is not a count of the union of the divisibility sets. The phrase exactly two requires removing every copy of the four triple-divisible integers rather than leaving one copy as in an at-least-two count.

Why not the others: (B) leaves triples counted three times; (C) removes each only once; (D) subtracts each four times.

Key point: Determine multiplicities for the specific event being counted.

Q33 / B equivalence relation test

Answer: (B) Four congruence classes.

Worked derivation:

$$a-a=0; \quad a-b=4k \Rightarrow b-a=-4k; \quad a-b=4k,\ b-c=4j \Rightarrow a-c=4(k+j).$$

Why: The three displayed checks establish reflexivity, symmetry and transitivity. Every integer has one residue 0,1,2,3 modulo 4, and precisely integers with equal residues are related. The four residue classes partition all integers because any integer has exactly one remainder on division by four.

Why not the others: (A) denies transitivity even though differences add; (C) denies reflexivity although a−a=0; (D) denies symmetry and transitivity despite signed and summed multiples of four.

Key point: Verify all three properties before counting classes.

Q34 / D roots-of-unity geometry

Answer: (D) 3√3/4.

Given: Each pair of consecutive roots subtends a central angle of 120°.

Formula/Principle: Chord length=2sin60°=√3; equilateral area=(√3/4)s².

Step 1: Substitute s²=3.

Worked derivation:

$$s=2\sin(\pi/3)=\sqrt{3}, \qquad A=\frac{\sqrt{3}}{4}(\sqrt{3})^2=\frac{3\sqrt{3}}{4}.$$

Why: The roots lie evenly spaced on a circle of radius one, so the side is the chord, not the radius. Using the distance from origin to a root as the triangle side would be wrong: the side connects two roots separated by 120°.

Why not the others: (A) doubles the area; (B) uses the wrong side; (C) uses radius as side.

Key point: Determine chord length before area.

Q35 / C substitution and bounds

Answer: (C) 8/3.

Given: Put u=4−x², so du=−2x dx and bounds x=0,2 map to u=4,0.

Formula/Principle: Reverse the bounds to absorb the minus sign, then integrate u¹ᐟ².

Step 1: The integral is ½∫₀⁴√u du.

Worked derivation:

$$\int_0^2 x\sqrt{4-x^2}\,dx=\frac{1}{2}\int_0^4 u^{1/2}du=\frac{1}{3}\left[u^{3/2}\right]_0^4=\frac{8}{3}.$$

Why: The original integrand is nonnegative, so reversing the transformed limits is necessary; omitting it would give a negative area. The resulting positive 8/3 also provides a sign check against the original nonnegative curve over the given interval.

Why not the others: (A) loses a factor two; (B) uses the endpoint radius alone; (D) doubles the result.

Key point: Transform the limits and differential together.

Q36 / A plane intersection

Answer: (A) (1,1,1).

Given: Subtract plane equations to get 2y=2.

Formula/Principle: The line consists of y=1 and x+z=2.

Step 1: For (1,1,1), y=1 and x+z=2.

Worked derivation:

$$(x+y+z)-(x-y+z)=2 \Rightarrow y=1, \quad x+z=2.$$

Why: Intersection membership requires simultaneous satisfaction, not just one plane. The first plane alone admits an entire plane of candidates; only its intersection with the second reduces them to a line.

Why not the others: (B) has x+z=1; (C) has y=0; (D) has y=2.

Key point: Check both constraints for each candidate.

Q37 / B monotonicity interval

Answer: (B) $a=1$.

Given: $f_a(x)=x^3-3ax$ on $x \geq 1$, with strict increase and minimum at most -2 required simultaneously.

Formula/Principle: Analyse the derivative on the full domain before imposing the minimum-value bound.

Step 1:

$$f_a'(x)=3(x^2-a).$$

If $a>1$, this derivative is negative just to the right of one, so global increase fails. If $a \leq 1$, it is positive for every $x>1$, giving strict increase.

Step 2: Under that condition the minimum is attained at the left endpoint:

$$\min f_a = f_a(1)=1-3a.$$

Step 3:

$$1-3a \leq -2 \Rightarrow a \geq 1.$$

Step 4: Intersecting $a \leq 1$ and $a \geq 1$ leaves only one; at $a=1$ the endpoint derivative is zero but every later derivative is positive.

Worked derivation:

$$a \leq 1, \quad a \geq 1 \Rightarrow a=1.$$

Why: A zero derivative at a single endpoint does not prevent strict increase on the whole half-line. The minimum bound must be evaluated only after monotonicity has established where the minimum occurs.

Why not the others: (A) violates local monotonicity near one; (C) includes minima above -2; (D) rejects a valid equality case.

Key point: Intersect global behaviour with the value condition, retaining valid endpoints.

Q38 / C standard-deviation constraint

Answer: (C) 6, inconsistent.

Given: Sum=3×4=12, so third=6; deviations from 4 are −2,0,+2.

Formula/Principle: Population variance=(sum squared deviations)/3.

Step 1: Variance=(4+0+4)/3=8/3, not 8.

Worked derivation:

$$x_3 = 12 - 2 - 4 = 6, \quad \sigma^2 = \frac{(2-4)^2 + (4-4)^2 + (6-4)^2}{3} = \frac{8}{3}.$$

Why: Mean constrains the missing datum, but the variance supplies a separate consistency check. The calculated 8/3 differs from the reported 8, so no third observation can satisfy both summaries simultaneously.

Why not the others: (A) uses the mean but fails the variance test; (B) chooses 8 as the datum and falsely accepts the variance; (D) also chooses 8 and rejects variance without first satisfying the mean.

Key point: Check both reported summary statistics.

Q39 / D removable singularity

Answer: (D) 6.

Given: $x^2-9=(x-3)(x+3)$.

Formula/Principle: Removable continuity requires $f(3)=\lim_{x\to 3} f(x)$.

Step 1: Cancel only for $x\neq 3$, then evaluate $x+3$ at 3.

Worked derivation:

$$\lim_{x\to 3} \frac{x^2-9}{x-3} = \lim_{x\to 3}(x+3) = 6.$$

Why: The algebra gives the limiting value, but the original quotient remains undefined at x=3 until f(3) is explicitly assigned. This assigned point does not change any neighboring values; it only removes the discontinuity at the missing input.

Why not the others: (A) substitutes into numerator only; (B) uses x; (C) uses x^2.

Key point: Fill a hole with its finite limit.

Q40 / C binomial probability event

Answer: (C) 3/8.

Given: With the first head fixed, four independent tosses remain.

Formula/Principle: Exactly two heads among four has C(4,2) favourable sequences out of 2^4.

Step 1: 6/16=3/8.

Worked derivation:

$$P(H_{\text{total}} = 3 \mid H_1) = \binom{4}{2}(1/2)^4 = \frac{3}{8}.$$

Why: Conditioning changes the sample space; the first head already contributes one of the required three. Among the remaining four tosses, exactly two heads and two tails are needed; that is why the combination count is six rather than ten.

Why not the others: (A) undercounts; (B) is the unconditional probability; (D) ignores the exact-count restriction.

Key point: Reduce both required successes and remaining trials.

Q41 / B trig equation endpoints

Answer: (B) 5π/2.

Given: Write sin x+cos x=√2 sin(x+π/4).

Formula/Principle: Solve sin(x+π/4)=1/√2 over a full closed period.

Step 1: x=0, π/2 and 2π are the distinct solutions in [0,2π].

Worked derivation:

$$\sum x = 0 + \frac{\pi}{2} + 2\pi = \frac{5\pi}{2}.$$

Why: Both endpoints represent different real x values and satisfy the equation, though their trig states are the same; a half-open interval would omit one. Summing roots rather than merely counting them makes the 2π endpoint contribute to the answer even though it shares a sine/cosine state with zero.

Why not the others: (A) omits 2π; (C) omits π/2; (D) loses one endpoint and adds an unrelated angle.

Key point: Respect the specified closed interval when summing roots.

Q42 / A first-order linear solution

Answer: (A) 7/6.

Given: Integrating factor for y′+y/x is x.

Formula/Principle: Multiply through by x to obtain (xy)′=x².

Step 1: xy=x³/3+C; at x=1, y=0 implies C=−1/3.

Worked derivation:

$$xy = \frac{x^3 - 1}{3} \Rightarrow y(2) = \frac{2^3 - 1}{3 \cdot 2} = \frac{7}{6}.$$

Why: The nonconstant coefficient changes the integrating factor; simply solving with a constant exponential transient would not satisfy the differential equation. Multiplying by x is valid because the domain excludes zero; it turns both left-hand terms into the derivative of xy.

Why not the others: (B) omits division by x; (C) loses the cubic contribution; (D) uses the wrong initial constant.

Key point: Check the integrating factor before integrating.

Q43 / D Root-square constraint and real positive roots

Answer: (D) $[1/2, 1]$.

Given: $\alpha + \beta = 2,\ \alpha\beta = k$.

Formula/Principle: Combine reality, positivity and the sum-of-squares constraint.

Step 1: Reality requires $4 - 4k \geq 0$, hence $k \leq 1$.

Step 2:

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 4 - 2k \leq 3 \Rightarrow k \geq \frac{1}{2}.$$

Step 3: On this interval product k is positive and sum is positive, so both real roots are positive. At $k = 1$ both equal one, which is allowed.

Worked derivation:

$$\{k : k \leq 1\} \cap \{k : k \geq 1/2\} = [1/2, 1].$$

Why: No single condition guarantees all three requirements; their intersection is essential. The endpoint at one must remain because the question explicitly permits equal roots.

Why not the others: (A) includes roots whose squared sum is too large; (B) includes nonreal roots for $k > 1$; (C) discards the valid repeated-root endpoint.

Key point: Translate each root condition separately before intersecting parameter ranges.

Q44 / C parabola normal

Answer: (C) −1.

Given: Differentiate y²=4x implicitly.

Formula/Principle: Tangent slope=2/y=1 at y=2; normal is negative reciprocal.

Step 1: Normal slope=−1.

Worked derivation:

$$2y\frac{dy}{dx} = 4 \Rightarrow m_t = \frac{2}{y} = 1, \quad m_n = -\frac{1}{m_t} = -1.$$

Why: A normal is perpendicular to the tangent, so use the negative reciprocal of its slope. Evaluating the tangent slope at the specified point is essential before taking its perpendicular direction; the symbolic slope varies along the parabola.

Why not the others: (A) omits the reciprocal after an incorrect tangent slope; (B) loses sign; (D) is tangent.

Key point: Differentiate first, then take negative reciprocal.

Q45 / B dot-product projection

Answer: (B) 4/3.

Given: a·b=2−2+4=4 and norm b=3.

Formula/Principle: Scalar projection of a on b is (a·b)/norm b.

Step 1: 4/3.

Worked derivation:

$$\text{comp}_b\, a = \frac{a \cdot b}{\|b\|} = \frac{4}{\sqrt{1+4+4}} = \frac{4}{3}.$$

Why: Dividing by norm b once gives the signed length along b; dividing by its square would give a vector coefficient instead. The projection is positive because the dot product is positive; using norm squared would instead give the coefficient in the vector projection formula.

Why not the others: (A) is dot product; (C) doubles it; (D) is norm b.

Key point: Scalar and vector projection use different denominators.

Physics

Q46 / A piecewise acceleration graph

Answer: (A) 3 m s⁻¹.

Given: Initial velocity 3; signed acceleration areas are 2×2 and −1×4.

Formula/Principle: Δv=∫a dt, signed area under a-t plot.

Step 1: Net Δv=4−4=0.

Worked derivation:

$$v(6) = 3 + (2)(2) + (-1)(4) = 3 \text{ m s}^{-1}.$$

Why: Opposite acceleration intervals cancel velocity change, even though distance travelled cannot be inferred from net area alone. The graph has a larger positive acceleration for a shorter interval and a smaller negative acceleration for longer; their areas, not their heights, cancel.

Why not the others: (B) ignores first interval; (C) drops initial velocity; (D) ignores negative area.

Key point: Use signed acceleration area.

Q47 / B spring-gravity conversion

Answer: (B) 0.20 m.

Given: Lost gravitational potential is mgh=2 J.

Formula/Principle: At maximum compression, kinetic energy is zero and ½kx²=mgh.

Step 1: 50x²=2 so x=0.20 m.

Worked derivation:

$$x = \sqrt{2mgh/k} = \sqrt{(2)(0.5)(10)(0.4)/100} = 0.20 \text{ m}.$$

Why: Energy is conserved across the descent and compression; taking the square root is required after equating potential energies. At maximum compression the block pauses, so no kinetic energy term remains; the horizontal travel was stipulated not to change gravitational potential.

Why not the others: (A) halves x; (C) equates height with compression; (D) misuses spring energy.

Key point: State the reference and final zero-speed condition.

Q48 / C torque versus angular acceleration

Answer: (C) 4.

Given: Equal net torque τ; second rotor has four times inertia.

Formula/Principle: τ=Iα.

Step 1: $\alpha_1=\tau/I$, $\alpha_2=\tau/(4I)$.

Worked derivation:

$$\frac{\alpha_1}{\alpha_2} = \frac{\tau/I}{\tau/(4I)} = 4.$$

Why: Moment of inertia resists angular acceleration for a specified torque, so multiplying inertia by four divides acceleration by four. This inverse dependence assumes the same net torque, which is why the ratio is four rather than a comparison of rotational kinetic energies.

Why not the others: (A) inverts ratio; (B) takes square root; (D) squares it.

Key point: Specify which rotor is numerator.

Q49 / D Buoyancy-dependent Stokes speed and dissipated power

Answer: (D) Speed ratio 2; dissipated-power ratio 8.

Given: The density differences from the fluid are 2ρ for A and ρ for B.

Formula/Principle: Balance effective weight with Stokes drag:

$$6\pi\eta r v = \frac{4}{3}\pi r^3(\rho_s - \rho)g.$$

Step 1: Thus $v \propto r^2(\rho_s - \rho)$, giving

$$v_B/v_A = 2^2(1/2) = 2.$$

Step 2: At terminal speed drag equals effective weight, so

$$F_B/F_A = 2^3(1/2) = 4.$$

Step 3:

$$P_B/P_A = \frac{F_B v_B}{F_A v_A} = 4(2) = 8.$$

Worked derivation:

$$v_B/v_A = 2, \qquad P_B/P_A = 8.$$

Why: Buoyancy must be subtracted before comparing the driving weights; the spheres do not have equal density contrasts. Power depends on both drag force and speed, so its ratio differs from either one separately.

Why not the others: (A) ignores the contrast change in both balances; (B) reports the force ratio as the power ratio; (C) combines an incorrect speed ratio with the power result.

Key point: Include buoyancy in terminal motion, then calculate force times speed.

Q50 / A adiabatic process inference

Answer: (A) Work draws from internal energy.

Worked derivation:

$$Q = 0, \quad \Delta U = Q - W = -W < 0 \quad (W > 0).$$

Why: Adiabatic means no heat exchange, not constant temperature. In a reversible expansion, the gas does positive work and therefore cools as its internal energy decreases.

Why not the others: (B) violates first law; (C) implies no expansion; (D) reverses work sign and adds heat.

Key point: Declare work sign convention.

Q51 / B beat ambiguity with reference change

Answer: (B) 254 Hz.

Given: First observation gives possible fork frequencies 246 or 254 Hz.

Formula/Principle: Beat frequency is the magnitude of frequency difference.

Step 1: Against 252 Hz, 254 yields 2 beats whereas 246 yields 6.

Worked derivation:

$$|f - 250| = 4 \Rightarrow f \in \{246, 254\}; \quad |f - 252| = 2 \Rightarrow f = 254.$$

Why: One beat rate alone has an above/below ambiguity; changing the reference resolves it by a second constraint. The second reference is closer to 254 Hz but farther from 246 Hz, so the observed beat reduction selects the higher candidate.

Why not the others: (A) yields 6 second beats; (C) fails first; (D) yields zero second beats.

Key point: Intersect both absolute-value conditions.

Q52 / C satellite period under radius change

Answer: (C) 8.

Given: Central mass and G remain fixed.

Formula/Principle: Circular orbit balance yields $T \propto r^{3/2}$.

Step 1: $(4r/r)^{3/2} = 4^{3/2} = 8$.

Worked derivation:

$$T = 2\pi\sqrt{\frac{r^3}{GM}}, \quad \frac{T_2}{T_1} = \sqrt{4^3} = 8.$$

Why: The orbiting speed also changes, so the period cannot be inferred from circumference ratio alone. Force balance gives speed proportional to $r^{-1/2}$, which combines with a circumference proportional to r to produce period proportional to $r^{3/2}$.

Why not the others: (A) square-root ratio; (B) circumference ratio; (D) squares radius ratio.

Key point: Use Kepler scaling derived from force balance.

Q53 / A electric potential from field region

Answer: (A) +E₀a.

Given: Signed field area is E₀a−2E₀a=−E₀a.

Formula/Principle: Potential change ΔV=−∫Ex dx.

Step 1: Negate the signed area to obtain +E₀a.

Worked derivation:

$$V(2a) - V(0) = -[E_0 a - 2E_0 a] = +E_0 a.$$

Why: A field pointing toward −x in the second region makes potential increase as x increases there; its larger magnitude dominates the first interval. The negative-field segment contributes twice the potential rise produced by the positive-field segment's drop, leaving a net positive change.

Why not the others: (B) forgets final negative; (C) adds magnitudes and reverses sign; (D) adds magnitudes.

Key point: Integrate the signed field then negate.

Q54 / D Mechanical and battery work during capacitor separation

Answer: (D) External work $+U/2$; battery work $-U$.

Given: $U = CV^2/2$; doubling the gap changes capacitance to $C/2$.

Formula/Principle: Battery work is $V\Delta Q$; energy balance is $\Delta U = W_{\text{external}} + W_{\text{battery}}$.

Step 1:

$$U_f = \frac{1}{2}(C/2)V^2 = U/2, \quad \Delta U = -U/2.$$

Step 2:

$$\Delta Q = (C/2)V - CV = -CV/2, \quad W_{\text{battery}} = -CV^2/2 = -U.$$

Step 3:

$$W_{\text{external}} = \Delta U - W_{\text{battery}} = -U/2 + U = +U/2.$$

Worked derivation:

$$-U/2 = (+U/2) + (-U).$$

Why: The battery absorbs returning charge as capacitance falls, so its work on the electrical system is negative. Separating the plates against attraction requires positive external work even though the field's stored energy decreases.

Why not the others: (A) reverses both work signs; (B) interchanges energy magnitudes; (C) ignores the mechanical work needed against attraction.

Key point: Distinguish field-energy change from battery and mechanical work.

Q55 / D internal resistance terminal voltage

Answer: (D) 10 V.

Given: Total series resistance 6 Ω.

Formula/Principle: Current I=E/(R+r), terminal voltage V=E−Ir=IR.

Step 1: I=2 A; V=12−2×1=10 V.

Worked derivation:

$$I = \frac{12}{5+1} = 2\ \text{A}, \quad V = IR = (2)(5) = 10\ \text{V}.$$

Why: Terminal voltage is lower than open-circuit emf when current flows because internal resistance consumes part of the emf. The 2 V internal drop is not the delivered voltage; the external 5 Ω resistor receives the remaining 10 V at 2 A.

Why not the others: (A) adds internal drop; (B) ignores internal loss; (C) is internal drop.

Key point: Use total resistance for current.

Q56 / B parallel-wire force

Answer: (B) Half, still attractive.

Worked derivation:

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d} \Rightarrow (F/L)_{2d} = \frac{1}{2}(F/L)_d.$$

Why: Same-direction currents attract; doubling only separation changes inverse-distance magnitude, not current directions or force sense. The attraction can also be seen from each wire's field acting on the other's current, with the separation entering the field magnitude inversely.

Why not the others: (A) reverses scaling; (C) uses square-law and changes sign; (D) ignores distance.

Key point: Magnetic line force scales as 1/d.

Q57 / C resonance bandwidth concept

Answer: (C) Lower, broader peak.

Worked derivation:

$$\omega_0 = \frac{1}{\sqrt{LC}}, \quad I_{\text{max}} = \frac{V}{R}, \quad Q = \frac{\omega_0 L}{R}.$$

Why: Increasing R lowers maximum current and quality factor; the reduced Q broadens the resonance around the same ideal LC frequency. The same driving voltage divided by a larger R gives less peak current, while greater damping spreads appreciable response over a wider band.

Why not the others: (A) reverses damping; (B) changes LC without changing either; (D) ignores finite R.

Key point: Resistance controls damping, while LC sets ideal resonance.

Q58 / B prism deviation

Answer: (B) 40°.

Given: i=e=50°, apex A=60°.

Formula/Principle: Prism deviation δ=i+e−A.

Step 1: 50+50−60=40°.

Worked derivation:

$$\delta = i + e - A = 50^\circ + 50^\circ - 60^\circ = 40^\circ.$$

Why: Symmetric passage gives equal entry and exit angles, but their sum must still be reduced by the prism apex. The sketch identifies angles relative to the normals.

Why not the others: (A) halves the incidence sum; (C) uses A/2; (D) equates deviation with incidence.

Key point: Label angles against surface normals.

Q59 / C spectrum wavelength and frequency

Answer: (C) 3/2.

Given: Both photons travel at c in vacuum.

Formula/Principle: f=c/λ, so frequency is inverse to wavelength.

Step 1: f400/f600=600/400=3/2.

Worked derivation:

$$\frac{f_{400}}{f_{600}} = \frac{c/(400\,\text{nm})}{c/(600\,\text{nm})} = \frac{600}{400} = \frac{3}{2}.$$

Why: The shorter wavelength photon has higher frequency; the wave speed cancels in the ratio. A factor 3/2 in frequency follows from the inverse ratio of wavelengths; photon energy would have that same ratio but is not squared.

Why not the others: (A) reverses ratio; (B) treats frequencies equal; (D) squares inverse-wavelength ratio.

Key point: Frequency is inversely proportional to wavelength.

Q60 / D half-life mixture

Answer: (D) 3/8.

Given: A retains 1/4 activity and B retains 1/2 after two hours.

Formula/Principle: Activities add; initial A_0=B_0.

Step 1: Final=(A_0/4+A_0/2)=3A_0/4; initial=2A_0.

Worked derivation:

$$\frac{A(2)+B(2)}{A(0)+B(0)} = \frac{A_0(1/4+1/2)}{2A_0} = \frac{3}{8}.$$

Why: Equal initial activities, not equal nuclei counts, set the weighting. Different half-lives make their remaining fractions unequal.

Why not the others: (A) uses A only; (B) uses B only; (C) sums fractions without correct initial normalization.

Key point: Weight decay fractions by initial activities.
